

Krishnendu Karmakar

Senior Backend Engineer | Golang | Cloud & Distributed Systems

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years of experience designing and scaling distributed systems in Golang. At Kaleyra (TATA Communications), built high-throughput APIs and microservices processing 8,000+ RPS, with deep expertise in gRPC, REST, GraphQL, RabbitMQ, AWS SQS, PostgreSQL, MongoDB, and Redis. Holds Executive PG Certification in Cloud Computing and DevOps from IIT Roorkee, covering AWS, Azure, Kubernetes, Docker, Terraform, and CI/CD. Actively uses AI-assisted engineering workflows including Cursor and Claude for accelerated development, agentic problem solving, and code review. Focused on building high-throughput, resilient backend systems at scale.

PROFESSIONAL EXPERIENCE

Kaleyra (TATA Communications Ltd.)

Senior Software Engineer · Jun 2022 – Present · Remote, India

Promoted from Software Engineering Professional to Senior Software Engineer

High-Throughput SMS Delivery Platform

- Architected and optimised a global SMS delivery system processing 8,000+ RPS for enterprise clients across 190+ countries.
- Built and maintained 10+ public REST APIs and 10+ internal gRPC APIs with strict validation, backward compatibility, and comprehensive error handling.
- Migrated RabbitMQ and Redis workers from PHP to Golang — achieved 90% reduction in message processing latency and significantly reduced infrastructure load.
- Designed and implemented a fixed-window rate limiter in Go, cutting API overload incidents by 40% and stabilising system performance under peak traffic.
- Developed automated daily and monthly message-cap features with integrated email notifications, directly reducing enterprise billing disputes.

Active-Active Disaster Recovery — Multi-Region AWS Architecture

- Contributed to dual-region AWS deployments across India (ap-south-1/2) and Southeast Asia (ap-southeast-1, ap-east-1), ensuring high availability and geographic redundancy with live traffic across all sites.
- Engineered a custom Redis Cache Moderator — a coordination layer maintaining Redis data consistency across geographically distributed deployments without Redis Enterprise, eliminating vendor dependency.
- Collaborated with database teams to implement bi-directional synchronisation across MySQL and MongoDB clusters ensuring consistency without service interruption.

Cross-Team API Integration and Inbound SMS Orchestration

- Led end-to-end integration of a critical third-party customer data API — exposing billing details, currency, country, user identity, and feature entitlements — into the core SMS platform, coordinating across multiple internal teams.
- Drove stakeholder alignment across engineering, product, and billing teams, defining integration contracts, edge case handling, and rollout strategy.
- Built a config-driven inbound SMS system supporting automated replies and dynamic workflows with zero-downtime deployments via decoupled processing logic.
- Developed 5 internal REST APIs and 5 RabbitMQ workers orchestrating emails, SMS, webhooks, and subscriptions based on configurable customer rules.
- Implemented real-time tracking and structured logging — reduced average debugging time by ~60% for customer-facing issues.

Multi-Channel OTP Verification System

- Built a fault-tolerant OTP service across SMS, Email, and Voice Call with retry and timeout mechanisms at every delivery layer.
- Replaced Redis with DynamoDB as primary state store for OTP verification, achieving eventual consistency and eliminating cache-layer failures under cross-region load.
- Integrated AWS SQS as a buffering layer between OTP generation and delivery services, improving throughput and isolating failure domains.
- Authored full Low-Level Design (LLD) documentation to support cross-team collaboration and future scalability planning.

Wipro Limited

Software Engineer · Dec 2020 – May 2022 · Remote, Bengaluru

Computer Assembly Management Platform — HP (Client Project)

- Developed GraphQL and gRPC APIs in Golang for real-time data access and inter-service communication, streamlining HP assembly workflows.
- Built JWT-based authentication and role-based access control by modelling and extending PostgreSQL schema across 3 core entities.
- Created a Golang CLI tool to import and validate Excel datasets into PostgreSQL — eliminated 70% of manual data entry effort.
- Achieved 80%+ unit test coverage across two core repositories, significantly reducing production regression incidents.
- Worked in agile sprints using JIRA and Azure Boards; actively participated in sprint planning and code reviews.

SKILLS & TECHNOLOGIES

Languages	Golang (primary) · Python · Rust (learning) · PHP
APIs & Protocols	REST · gRPC · GraphQL · WebSockets
Databases	PostgreSQL · MySQL · MongoDB · Redis · DynamoDB
Message Brokers	RabbitMQ · AWS SQS · Redis Queue
Cloud & DevOps	AWS (SQS, EC2, S3, IAM, VPC, ELB, DynamoDB) · Azure · Docker · Kubernetes · Ansible · Jenkins · CI/CD · Terraform · Linux
Architecture	Microservices · Event-Driven Design · Rate Limiting · API Gateway · LLD/HLD · Distributed Systems
AI & Dev Tools	Cursor · Claude · GitHub Copilot · AI-assisted development · Git · VSCode · Insomnia · JIRA · Draw.io

KEY PROJECTS (OPEN SOURCE)

Kube-Pulse (Active) | Go · Rust · gRPC · Kubernetes

Microservices-based Kubernetes observability system: gateway (Go/REST), collector (Rust/K8s metrics), and analyser (Rust/anomaly detection). Services communicate over gRPC.

Football Transfers Recorder | Go · PostgreSQL · Docker · sqlc · JWT

REST API to record and track football transfer data with JWT auth, GitHub Actions CI/CD, and dockerised deployment.

CODING PROFILES

Exercism	exercism.org/profiles/krish8learn — Practice tracks across multiple languages
HackerRank	hackerrank.com/profile/krishkarmakar17 — SQL and algorithmic problem sets
Codility	github.com/krish8learn/Codility_Training — Codility problem solutions
Data Structures	github.com/krish8learn/DataStructures — Hands-on implementations of core data structures
Technical Writing	medium.com/@krishkarmakar17 — Articles on software engineering and distributed systems

EDUCATION

B.Tech — Electronics and Communication Engineering

Kalyani Government Engineering College, MAKAUT University · CGPA: 8.5 · 2017–2020

Diploma — Electronics and Telecommunication Engineering

Jnan Chandra Ghosh Polytechnic, WBSCT & VE & SD · CGPA: 8.6 · 2014–2017

CERTIFICATIONS

- Executive PG Certification — Cloud Computing and DevOps | IIT Roorkee / Intellipaat · Mar 2026
- Rust Fundamentals | Duke University / Coursera · Apr 2026
- Golang Master Certification | Udemy · Jan 2021
- PostgreSQL Certification | Udemy · Feb 2021
- Python Certification | Coursera / University of Michigan · Sep 2020
- MongoDB Basic Course | MongoDB University · Mar 2022

LANGUAGES

Bengali (Native) · English (Fluent) · Hindi (Fluent) · German (Basic / A2 — actively learning)